

Perspectives on AI, Business and Sustainability PAIBS

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Term 1 (October 8th - December 3rd) → *From challenge to solution concept*

Definition and Solution Design - identifying meaningful challenges, understanding stakeholder needs, and proposing comprehensive solution.

Term 2 (February 12nd - May 14th) → *From concept to working prototype*

Technical Implementation - technical development, culminating in a fully developed prototype and business case.

Term 2		
Date	Content	Teacher/s
February 12, 2026	Literature Replication + Recap	Marc Herrera
February 19, 2026	Prototyping tools and techniques	Jose Rodriguez
March 12, 2026	Front-end prototyping	Jose Rodriguez
April 9, 2026	Prototyping AI applications	Jose Rodriguez
April 16, 2026	Interpreting AI-Based EO Results and Value	Marc Herrera
April 30, 2026	Impact assessment	Angelo Romasanta
May 14, 2026	Final Presentations	Marc Herrera / Angelo Romasanta

1. Select a **topic/use case** and define the **problem** to solve.

Considerations:

- Scope → “What does it cover?”
- Impact → “Why does it matter?”
- Vision → “Where is it going?”
- Objectives → “What will be achieved?”
- Limitations → “What can/not be done?”
- Business → “Who will use or benefit from it?”

Term 1: Definition

2. Define the **initial boundary conditions**: temporal and spatial extent, other conditions.
3. **Data available**, both EO and Ground Reference. Open Access, preprocess required.
4. Define the **final boundary conditions**: temporal and spatial extent, other conditions.

Plan and account for everything from the start

Term 2: Implementation

What Next? December → February

Technical

- 1.** Data Obtention.
 - a. All Data in Detail Definition: Spatial & Temporal Extent, Sources, GT, Auxiliar.
 - b. Data Downloading.
 - c. Data Loading and Initial Analysis.

- 2.** Initial AI Models Brainstorming.
 - a. ML vs DL architectures and algorithms.
 - b. Interpretable/Explainable AI.
 - c. Scalability and Implementability.

What Next? December → February

Idea, Product and Value

1. Clear Final Product + Added Value.
 - a. What problem are we solving?
 - b. What is our contribution, and how do we differentiate ourselves from existing solutions or competitors?
 - c. Who is the end user or client?

2. Business Model → Where is the revenue coming from.

3. **Contact Stakeholders:** Public/Private Institutions, Research Centers, Companies, Governments.
 - a. Auxiliar data provider or Experts.
 - b. End users or clients.